Section 1.6: Euler's Theorem and Public Key Cryptography

Tuan Tran

A motivating question

Problem

Compute $7^{100} \pmod{15}$.

- Direct computation is unrealistic.
- Even repeated squaring feels blind if we do not know what to expect.
- We need a general principle for powers modulo n.

When do powers return to 1 modulo n?

Why should a pattern exist?

Consider powers of 2 modulo 7:

$$1,\ 2,\ 4,\ 8 \equiv 1,\ 2,\ 4,\ 1, \ldots$$

- Only finitely many congruence classes modulo 7 exist.
- Powers cannot keep producing new residues forever.
- So some repetition must occur.

The real question is:

Can we predict the exponent where this repetition appears?

A first restriction: not every residue behaves well

If $a^k \equiv 1 \pmod{n}$ for some positive integer k, then $\gcd(a, n) = 1$. (Since $a \cdot a^{k-1} \equiv 1 \pmod{n}$, a has a multiplicative inverse modulo n, and hence $\gcd(a, n) = 1$.)

For a fixed modulus n, the integers a with $\gcd(a, n) = 1$ have special properties.

- They are exactly the congruence classes that have multiplicative inverses modulo n.
- Cancellation law: if $ax \equiv ay \pmod{n}$, then $x \equiv y \pmod{n}$.
- Euler's theorem concerns precisely these residues.

Key idea

To understand powers modulo n, we first count how many integers are coprime to n.

Definition of Euler's totient function

Definition

For a positive integer n, define

$$\varphi(n) = \left|\{1 \leq a \leq n : \gcd(a, n) = 1\}\right|.$$

In words, $\varphi(n)$ is the number of integers from 1 to n that are coprime to n.

Examples:

$$\varphi(10) = 4, \qquad \varphi(12) = 4, \qquad \varphi(15) = 8.$$

Computing $\varphi(12)$ directly

The numbers from 1 to 12 are

$$1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12.$$

Those not coprime to 12 are exactly the multiples of 2 or 3.

So the numbers coprime to 12 are

$$1, 5, 7, 11.$$

Hence

$$\varphi(12) = 4.$$

We are really counting integers that avoid the prime divisors of 12.

The prime case

Suppose p is prime.

Among the integers $1, 2, \ldots, p$, the only one not coprime to p is p itself.

Therefore

$$\varphi(p) = p - 1.$$

Examples:

$$\varphi(2) = 1, \qquad \varphi(5) = 4, \qquad \varphi(13) = 12.$$

This will later give Fermat's little theorem as a special case of Euler's theorem.

Prime powers

Now let $n = p^k$, where p is prime and $k \geq 1$.

- There are p^k integers from 1 to p^k.
- An integer fails to be coprime to p^k exactly when it is divisible by p.
- The multiples of p in this list are

$$p, 2p, 3p, \ldots, p^k,$$

  so there are p^{k-1} of them.

Hence

$$\varphi(p^k) = p^k - p^{k-1} = p^k\left(1 - \frac{1}{p}\right).$$

The case of two distinct primes

Let $n = pq$, where p and q are distinct primes.

To count numbers coprime to pq, remove those divisible by p or by q.

- Total number of integers from 1 to pq: pq.
- Multiples of p: q.
- Multiples of q: p.
- Multiples of both: only pq itself.

So, by inclusion–exclusion,

$$\varphi(pq) = pq - q - p + 1 = (p-1)(q-1).$$

The general formula for $\varphi(n)$

If

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_r^{\alpha_r}$$

is the prime factorization of n, then

$$\boxed{\varphi(n) = n \prod_{i=1}^{r} \left(1 - \frac{1}{p_i}\right).}$$

- Only the *distinct* prime divisors matter.
- To be coprime to n, an integer must avoid divisibility by each prime p_i.
- This formula is what makes Euler's theorem usable in practice.

Example:

$$\varphi(20) = 20\left(1 - \frac{1}{2}\right)\left(1 - \frac{1}{5}\right) = 8.$$

Euler's theorem

Theorem

If $\gcd(a, n) = 1$, then
$$a^{\varphi(n)} \equiv 1 \pmod{n}.$$

This is the main theorem of the section.

- It applies to any positive integer n.
- The only requirement is that a be coprime to n.
- It turns a large power into something very manageable.

Why should this be true?

Let $x_1, \ldots, x_{\varphi(n)}$ be all the integers in $\{1, 2, \ldots, n\}$ that are coprime to n.

Since $\gcd(a, n) = 1$, multiplying by a preserves coprimeness with n.

So

$$ax_1,\ ax_2,\ \ldots,\ ax_{\varphi(n)}$$

are also all coprime to n modulo n.

The key point is that these give the *same set of residues*, just in a different order.

The key permutation idea

Why do we get the same set again?

Suppose

$$ax_i \equiv ax_j \pmod{n}.$$

Then

$$a(x_i - x_j) \equiv 0 \pmod{n}.$$

Because $\gcd(a, n) = 1$, by the cancellation law, we have

$$x_i \equiv x_j \pmod{n}.$$

Since both lie between 1 and n, this gives $x_i = x_j$.

Therefore multiplication by a simply rearranges the reduced residue system.

Proof of Euler's theorem

Since $ax_1, \ldots, ax_{\varphi(n)}$ are the same residues as $x_1, \ldots, x_{\varphi(n)}$, we have

$$(ax_1)(ax_2)\cdots(ax_{\varphi(n)}) \equiv x_1x_2\cdots x_{\varphi(n)} \pmod{n}.$$

So

$$a^{\varphi(n)}x_1x_2\cdots x_{\varphi(n)} \equiv x_1x_2\cdots x_{\varphi(n)} \pmod{n}.$$

Each x_i is coprime to n, so the product $x_1x_2\cdots x_{\varphi(n)}$ is also coprime to n. Hence we may cancel it modulo n and obtain

$$a^{\varphi(n)} \equiv 1 \pmod{n}.$$

A first example

Compute $3^{100} \pmod{10}$.

Since $(3, 10) = 1$ and

$$\varphi(10) = 10\Big(1 - \frac{1}{2}\Big)\Big(1 - \frac{1}{5}\Big) = 4,$$

Euler's theorem gives

$$3^4 \equiv 1 \pmod{10}.$$

Now

$$3^{100} = (3^4)^{25} \equiv 1^{25} \equiv 1 \pmod{10}.$$

So the last digit of 3^{100} is 1.

A second example

Compute $7^{222} \pmod{15}$.

Since $(7, 15) = 1$ and

$$\varphi(15) = 15\Big(1 - \frac{1}{3}\Big)\Big(1 - \frac{1}{5}\Big) = 8,$$

Euler's theorem gives

$$7^8 \equiv 1 \pmod{15}.$$

Because

$$222 = 8 \cdot 27 + 6,$$

we get

$$7^{222} = (7^8)^{27} 7^6 \equiv 7^6 \pmod{15}.$$

Finishing the computation

We continue from

$$7^{222} \equiv 7^6 \pmod{15}.$$

Now

$$7^2 = 49 \equiv 4 \pmod{15},$$

so

$$7^4 \equiv 4^2 = 16 \equiv 1 \pmod{15}.$$

Therefore

$$7^6 = 7^4 \cdot 7^2 \equiv 1 \cdot 4 \equiv 4 \pmod{15}.$$

Hence

$$\boxed{7^{222} \equiv 4 \pmod{15}.}$$

Fermat's little theorem as a special case

If p is prime and $p \nmid a$, then

$$\varphi(p) = p - 1.$$

Applying Euler's theorem with $n = p$, we obtain

$$a^{p-1} \equiv 1 \pmod{p}.$$

This is Fermat's little theorem.

A useful consequence: the Euler formula for inverses

Suppose $\gcd(a, n) = 1$. Since

$$a^{\varphi(n)} \equiv 1 \pmod{n},$$

we may rewrite this as

$$a \cdot a^{\varphi(n)-1} \equiv 1 \pmod{n}.$$

Therefore

$$\boxed{a^{-1} \equiv a^{\varphi(n)-1} \pmod{n}.}$$

In words: Euler's theorem gives a formula for the multiplicative inverse of a modulo n.

Example of the inverse formula

Find the inverse of 3 modulo 10.

Since $(3, 10) = 1$ and $\varphi(10) = 4$, Euler's formula gives

$$3^{-1} \equiv 3^{\varphi(10)-1} = 3^3 = 27 \equiv 7 \pmod{10}.$$

Check:

$$3 \cdot 7 = 21 \equiv 1 \pmod{10}.$$

This idea will be very useful in the next part, when we discuss public key cryptography.

The problem of secret communication

Suppose Alice wants to send Bob a secret message.

A simple idea is to use a common secret key, but then a new problem appears:

How can Alice and Bob agree on the key safely?

If the key is sent through a public channel, anyone can read it.

Public key cryptography solves this by using *two* keys instead of one.

The basic idea of public key cryptography

Each user has two keys:

- a public key, which everyone may know,
- a private key, which only the owner knows.

The public key is used for encryption.

The private key is used for decryption.

So anyone can send Bob an encrypted message, but only Bob can recover it.

The RSA cryptosystem

The RSA construction uses almost every major idea from Chapter 1:

- prime numbers,
- greatest common divisors,
- the Euclidean algorithm,
- modular inverses,
- Euler's totient function,
- Euler's theorem (the mathematical engine behind RSA).

This is a very nice example of pure mathematics leading to a striking application.

Step 1: choose two primes

Bob begins by choosing two distinct primes p and q.

Then he forms

$$n = pq.$$

Because p and q are prime, we know

$$\varphi(n) = \varphi(pq) = (p-1)(q-1).$$

This is the key quantity in the construction.

Step 2: choose an exponent e

Next, Bob chooses an integer e such that

$$1 < e < \varphi(n) \qquad \text{and} \qquad \gcd(e, \varphi(n)) = 1.$$

Why do we need this?

- Because then e has an inverse modulo $\varphi(n)$.
- That inverse will become the private key.

Step 3: compute the inverse of e

Since $\gcd(e, \varphi(n)) = 1$, there exists d such that

$$ed \equiv 1 \pmod{\varphi(n)}.$$

Equivalently, there is an integer k with

$$ed = 1 + k\varphi(n).$$

We can find d using the Euclidean algorithm.

Public key and private key

At this point Bob has:

- the modulus $n = pq$,
- the encryption exponent e,
- the decryption exponent d.

He publishes:

$$\boxed{(n, e)}$$

This is the public key.

He keeps secret:

$$\boxed{d}$$

This is the essential part of the private key.

How encryption works

Suppose Alice wants to send a message m.

We assume that m is represented by an integer with

$$0 \leq m < n.$$

Using Bob's public key (n, e), Alice computes the ciphertext

$$c \equiv m^e \pmod{n}.$$

She sends c publicly.

How decryption works

Bob receives the ciphertext c and computes

$$m' \equiv c^d \pmod{n}.$$

Since $c \equiv m^e \pmod{n}$, this becomes

$$m' \equiv (m^e)^d = m^{ed} \pmod{n}.$$

The crucial question is:

Why does this recover the original message m?

Why RSA works?

Recall that $ed \equiv 1 \pmod{\varphi(n)}$. So for some integer k,

$$ed = 1 + k\varphi(n).$$

If $\gcd(m, n) = 1$, then Euler's theorem gives

$$m^{\varphi(n)} \equiv 1 \pmod{n}.$$

Therefore

$$m' \equiv m^{ed} = m^{1+k\varphi(n)} = m\left(m^{\varphi(n)}\right)^k \equiv m \pmod{n}.$$

So decryption really gives back the original message.

A small example: setup

Let us choose

$$p = 5, \qquad q = 11.$$

Then

$$n = pq = 55, \qquad \varphi(55) = (5-1)(11-1) = 40.$$

Choose

$$e = 3.$$

Since $\gcd(3, 40) = 1$, we may compute its inverse modulo 40.

A small example: finding d

We need to solve
$$3d \equiv 1 \pmod{40}.$$
Since
$$3 \cdot 27 = 81 = 1 + 2 \cdot 40,$$
we may take
$$d = 27.$$

So the keys are:
$$\text{public key } (55, 3), \qquad \text{private key } 27.$$

A complete small example: encryption

Suppose Alice wants to send the message

$$m = 7.$$

Using the public key $(55, 3)$, she computes

$$c \equiv 7^3 \pmod{55}.$$

Now

$$7^3 = 343 = 6 \cdot 55 + 13,$$

so

$$\boxed{c = 13.}$$

Alice sends 13.

A small example: decryption

Bob receives $c = 13$ and computes

$$13^{27} \pmod{55}.$$

We simplify step by step:

$$\begin{aligned}
13^2 &\equiv 169 \equiv 4 \pmod{55},\\
13^4 &\equiv 4^2 = 16 \pmod{55},\\
13^8 &\equiv 16^2 = 256 \equiv 36 \pmod{55},\\
13^{16} &\equiv 36^2 = 1296 \equiv 31 \pmod{55}.
\end{aligned}$$

Finishing the decryption

Since

$$27 = 16 + 8 + 2 + 1,$$

we get

$$13^{27} = 13^{16} \cdot 13^{8} \cdot 13^{2} \cdot 13.$$

Using the previous slide,

$$13^{27} \equiv 31 \cdot 36 \cdot 4 \cdot 13 \pmod{55}.$$

Now

$$31 \cdot 36 = 1116 \equiv 16, \qquad 16 \cdot 4 \cdot 13 = 832 \equiv 7 \pmod{55}.$$

Hence Bob recovers

$$\boxed{m = 7.}$$

Why is RSA believed to be secure?

The public key reveals $n = pq$ and e.

To compute the private exponent d, one needs $\varphi(n)$. But to find $\varphi(n) = (p-1)(q-1)$, one needs the factorization of n.

So the security idea is:

multiplying two large primes is easy, but factoring their product is hard.

Summary of Chapter 1

Chapter 1 has developed the basic arithmetic of the integers and congruences.

Its main themes are:

- division with remainder,
- greatest common divisors,
- prime factorization,
- congruence classes,
- solving linear congruences,
- Euler's theorem and public key cryptography.

This section is the payoff: many earlier ideas now come together in one application.

Tool 1: division and greatest common divisors

The first key ideas were:

- the Euclidean algorithm,
- writing the gcd as a linear combination.

These let us answer questions such as:

- Does a divide b?
- Can we solve $ax + by = d$?

This is the computational foundation for modular inverses.

Tool 2: congruences and linear congruences

Then we learned how to work modulo n.

Typical problems are:

- solve $ax \equiv b \pmod{n}$,
- find inverses modulo n.

A particularly important fact is:

$$ax \equiv 1 \pmod{n}$$

has a solution exactly when $\gcd(a, n) = 1$.

So gcd information controls invertibility modulo n.

Tool 3: Euler's totient function and Euler's theorem

For each positive integer n, the totient function counts the invertible residue classes:

$$\varphi(n) = \big|\{1 \leq a \leq n : (a, n) = 1\}\big|.$$

If

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_r^{\alpha_r},$$

then

$$\varphi(n) = n \prod_{i=1}^{r} \Big(1 - \frac{1}{p_i}\Big).$$

Euler's theorem says that if $\gcd(a, n) = 1$, then

$$a^{\varphi(n)} \equiv 1 \pmod{n}.$$

The logical flow of Chapter 1

Division algorithm

$$\Downarrow$$

gcd and Euclidean algorithm

$$\Downarrow$$

Congruences and modular inverses

$$\Downarrow$$

Euler's theorem $\Longrightarrow$ RSA